

INFO 300: Honors Research Paper

Introduction

In high-pressure environments, the ability to focus can be the difference between success and failure. Whether it's a surgeon in an operating room, a pilot mid-flight, or a student participating in a business case competition, the demands on sustained attention are both intense and unforgiving. Attention is a phenomenon that changes from moment to moment, shaping how individuals process and respond to their environment. Specifically, sustained attention can be defined as the ability to maintain concentration over an extended period while filtering out irrelevant information (Huang et al., 2023). This ability is critical in both everyday tasks and high-pressure environments, yet it is not fixed — various neurocognitive factors contribute to natural fluctuations in attention, influencing the ability to stay “in the zone” or move away from the optimal attentional state (Esterman & Rothlein, 2019).

By pairing this understanding of sustained attention with real-world examples of its application, we can develop a broader perspective on what enables certain individuals to cultivate this skill. This paper seeks to examine the psychological and broader scientific background of sustained attention, the extent to which it can be trained, and its potential relevance to business case competitions. To this effect, this paper will propose a study designed to investigate how training for sustained attention might impact performance in business case competitions. Given that these competitions require participants to process large amounts of information, maintain focus under time constraints, and think critically under pressure, they provide a unique setting to explore the practical implications of attention training. The overarching goal is to gain insight into the mechanisms of sustained attention, explore methods

for its enhancement, and outline a study proposal that seeks to examine its effectiveness in a competitive, real-world context.

Background Research

With the complexity of cognitive function and the underlying processes, sustained attention is convoluted yet sparingly explored in attention research. Work on the subject has revealed a variety of cognitive functions that influence attention and its duration across various real-world settings. Research, more generally speaking, on the subject of sustained attention has been sparse in comparison to other aspects of attention like shifting, dividing, and attentional selection (Esterman & Rothlein, 2019). Esterman and Rothlein (2019) talk about the models of sustained attention. They discuss how there is a key differentiating factor for the investigations of sustained attention because it focuses on performing a single task over time to explain both the fluctuation within an individual and the individual differences in the overall ability to maintain stable task performance. A key nuance of the framework highlighted in this paper is that because multiple cognitive processes are relevant to sustained attention, the underlying neural architecture concerning the subject is significantly dense.

The intentionality of mind wandering is a key factor when it comes to the allocation of attention and potentially developing control over it through training. Esterman and Rothlein (2019) reveal that intentional mind-wandering decreases with task difficulty. Furthermore, motivation can also impact the ratios of intentional versus unintentional wandering. Van Vleet et al. (2016) found that two weeks of sustained attention training in older adults led to improved working memory span and verbal fluency, this reaffirms the author's point regarding the

underlying cognitive functions of sustained attention. The findings highlight that the training intended to test for sustained attention should not just target prolonging the state of “in the zone” but also the ability to recover from attentional lapse and mind wandering.

The capability of sustaining attention varies naturally among individuals, but research in the space also reveals that specific networks can be targeted with training to boost efficacy and impact. Fortenbaugh et al. (2017) focus on the moment-to-moment fluctuations in sustained attention, and how the neurocognitive systems of humans change over the lifespan and in different clinical populations. Additionally, Fortenbaugh et al. (2017) reveal that the use of a statistical approach as well as neuroimaging paradigms has allowed for the better characterization of neural networks and cognitive processes that support sustained attention. To take this a step further, the better characterization of the neural networks has allowed researchers to highlight the dynamic interactions within and across multiple networks that can predict the behavioral performance of individuals. This aspect is more relevant to clinical populations looking to study the specific challenges faced by individuals in sustaining attention with implications for clinical diagnoses.

Subjective experiences and the consequential attentional lapse are another area of study exploring task engagement and disengagement during sustained attention. Smallwood et al. (2004) conducted three experiments to explore this relationship. The experimental method employed two measures of subjective experience (thought probes and questionnaires) to examine how differences in awareness correspond to variations in both task performance (reaction time and errors) and psycho-physiological measures (heart rate and galvanic skin response). Humans can distinguish between the two components of subjective experience during a period of sustained attention: First, task-unrelated thoughts that correspond to an absent-minded

disengagement from the task, and second, the intentional instruction for the mind to direct attentional resources by thinking about the task performances beforehand in an attempt to help exceed one's natural ability to perform the task. These results help contextualize the efficacy of training to help prolong the sustenance of attention.

The insights on sustained attention can be categorized into three key takeaways spanning the process, mental fluctuations, and trainability. The first takeaway is that sustained attention is a dynamic process that results from a complex network of cognitive processes. These processes also contribute to the intentionality of mind wandering and how it varies with task complexity. The second takeaway is that sustained attention requires an understanding of moment-to-moment fluctuations. The study of these fluctuations in a cognitive context allows the determination of behavioral patterns associated with attentional lapses and their application to specific populations. The third takeaway is the relativity of these cognitive factors to the ability of the human brain to be trained to identify these circumstances by focusing the training on cognitive processes that are most deterministic for sustained attention.

Real World Examples

Sustained attention is a critical skill required to perform at high-stake jobs like surgeons and pilots, these roles require individuals to operate at maximum cognitive capacity while experiencing sub-optimal conditions for recuperation like hours of sleep and rest. With the stakes associated with these jobs, a fluctuation in attention can have serious or even fatal consequences. Understanding how sustained attention works in these intense real-world scenarios can provide helpful insight into the potential factors that help elongate the attention spans of individuals practicing these professions.

Pilots

The job of a pilot requires long hours and sustained attention to successfully operate the flight specifically during key operations like takeoff, landing, and handling any emergencies while en route. Ghaderi and Saghafi (2024) examined the efficacy of pilot vigilance assessment and studied the role of flight data and continuous performance testing in detecting random attention loss. The study identifies the Continuous Matching Task (CMT) and other Continuous Performance Test (CPT) variations as effective measures for sustained attention, processing speed, and alertness in an aviation context. CPT is a widely used neuropsychological test that measures sustained attention and vigilance; it requires subjects to monitor a stimulus presentation series (e.g., visually presented letters or auditory tones) for a specified target for 3–15 min. This necessitates the test subjects to retain a state of “attentiveness” or remain “in the zone” for an extended period of time. The study also highlighted that pilots experience attention lapses during more monotonous periods of tasks, for instance on long-haul flights when fatigue accumulates both cognitively and physically. This insight is important because it affirms the importance of attention training to avoid any errors caused due to attentional lapses or inconsistencies in performance, and the steps that the aviation industry is taking to ensure sustained attention for pilots.

Surgeons

Similar to pilots, the role and attention to detail necessitated by surgeons make sustained attention an integral skill for both the safety and success of the work carried out by them. Howie et al. (2024) experiments with the feasibility of wearable sensors to assess the cognitive load during clinical performance was tested, and a part of this study was the monitoring of the

surgeon's sustained attention using EEG (Electroencephalogram) based devices (Howie et al., 2024). The cognitive load faced by surgeons varies significantly when conducting the procedure and monitoring of them can help improve performance. The insights can be used to back up the importance of sustained attention training for surgeons and the impact it can have on their performance.

Case Competitions and Sustained Attention

Business case competitions are fast-paced and high-pressure events where teams of four to five students are given a complex business problem — typically in the form of a written document that introduces them to a real-world problem facing a company or organization. Teams are then tasked with understanding the charge, analyzing the issue, developing recommendations, and presenting the solution while being placed under strict time constraints. The typical time frames for a case range between 8, 12, 24, or 48 hours with the 24 hours being the most common scenario. In the allotted time, a team is expected to find a solution while having limited access to resources and external guidance. With the challenges posed by this task, the teams need to demonstrate increased productivity, long working hours, and improved attention to excel at the problem-solving process.

While the consequences of attentional lapses are not as severe as the ones faced in the aviation or medical professions, case competitions pose the challenge of sustaining attention and remaining “in the zone” while dealing with limited breaks, continuous mental effort, and impacted sleep. The tasks involved in a case competition like data analysis, researching, brainstorming, and decision-making require attention to detail and significant cognitive load. Taking our understanding of sustained attention from literature review and real-world examples

like those of pilots and surgeons, case competitions can be used as an environment to test the impact and effectiveness of training in improving competitor focus and optimizing problem-solving under time pressure.

Future Directions

Given the intellectually demanding and time-intensive nature of business case competitions, this paper proposes a small-scale exploratory study to test whether sustained attention training can improve focus and performance in this setting. Taking inspiration from Asati and Miyachi (2019), the study will use a combination of methods to assess the impact of attention training on participants.

A potential structure of the study is as follows:

- Sample: Convenience sampling of case competition participants (16–24 students)
- Random Assignment: Participants will be randomly assigned to either a control group or a treatment group (which receives attention training)
- Pre-Treatment Interviews: All participants will be interviewed about their usual strategies for maintaining focus and managing time during competitions
- Training Period: The treatment group will undergo sustained attention training for two weeks
- Competition Phase: Both groups will participate in a business case competition following the training period
- Post-Competition Interviews: After the event, all participants will be interviewed to reflect on perceived and observed changes in their attention and performance

While the study is intentionally small and simplified, the goal is to begin measuring the possible impact of attention training—commonly studied in fields like surgery and aviation—in academic, competitive contexts. Ultimately, the study aims to explore whether such training can support students in retaining focus over long periods and improving overall productivity.

Works Cited

- Asati, M., & Miyachi, T. (2019, July 10). *A short virtual reality mindfulness meditation training for regaining sustained attention*. arXiv.org. <https://arxiv.org/abs/1907.04487>
- Esterman, M., & Rothlein, D. (2019). Models of sustained attention. *Current Opinion in Psychology*, 174–180. <https://www.bu.edu/ballab/pubs/Esterman2019.pdf>
- Fortenbaugh, F. C., DeGutis, J., & Esterman, M. (2017). Recent theoretical, neural, and clinical advances in sustained attention research. *Annals of the New York Academy of Sciences*, 1396(1), 70–91. <https://doi.org/10.1111/nyas.13318>
- Ghaderi, A., & Saghafi, F. (2024). Enhancing pilot vigilance assessment: The role of flight data and continuous performance test in detecting random attention loss in short IFR flights. *Journal of Air Transport Management*, 120, <https://doi.org/10.1016/j.jairtraman.2024.102673>
- Huang, H., Li, R., & Zhang, J. (2023). A review of visual sustained attention: neural mechanisms and computational models. *PeerJ*, 11, e15351. <https://doi.org/10.7717/peerj.15351>
- Howie, E. E., Harari, R., Dias, R. D., Wigmore, S. J., Skipworth, R. J. E., & Yule, S. (2024). Feasibility of wearable sensors to assess cognitive load during clinical performance: Lessons learned and blueprint for success. *Journal of Surgical Research*, 302, 222–231. <https://doi.org/10.1016/j.jss.2024.07.009>
- Smallwood, J., a, Davies, J. B., Heim, D., Finnigan, F., Sudberry, M., O'Connor, R., Obonsawin, M., The Psychology Department, Glasgow Caledonian University, Glasgow, UK, Department of Psychology, The University of Strathclyde, Glasgow, UK, & Department of Psychology, The University of Stirling, Stirling, UK. (2004). Subjective experience and the attentional lapse: Task engagement and disengagement during sustained attention.

In *Consciousness and Cognition* (Vol. 13, pp. 657–690) [Journal-article].

<https://doi.org/10.1016/j.concog.2004.06.003>